

Scenario	$c_{U,X}$	$c_{U,Z}$	c_{X,W_1}	c_{X,W_2}	Interpretation	Expected bias
S_1	0	0	0	0	All deactivated	No bias expected (direct only)
S_2	+ext	+ext	0	0	Strong confounding on X and Z through U	Potentially high spurious bias
S_3	-ext	-ext	0	0	Reverse conf. favoring the unprivileged group	Spurious bias
S_4	+ext	0	0	0	Confounding on X favoring the privileged group	Spurious bias
S_5	0	+ext	0	0	Confounding on Z favoring the privileged group	Spurious bias via Z
S_6	-ext	0	0	0	Reverse confounding on X favoring the unprivileged group	Spurious bias
S_7	0	-ext	0	0	Reverse confounding on Z favoring the unprivileged group	Spurious bias via Z
S_8	0	0	+ext	+ext	Full indirect effects via W_1 and W_2	Indirect bias through both mediators
S_9	0	0	-ext	-ext	Complete inversion, strongly favoring the privileged group	Reversed bias (may result in reverse discrimination)
S_{10}	0	0	+ext	0	Effect of X limited only to education	Indirect bias affecting Y via W_1
S_{11}	0	0	0	+ext	Effect of X limited only to employment	Indirect bias affecting Y via W_2
S_{12}	0	0	-ext	0	Reverse discrimination on education favouring the unprivileged group	Reversed bias via W_1
S_{13}	0	0	0	-ext	Reverse discrimination on employment favouring the unprivileged group	Reversed bias via W_2
S_{14}	+ext	+ext	+ext	+ext	All components fully activated	Potentially high total bias

Table 13: All Scenarios for the M_G example